

Trajectory Tracking for The Magnetic Ball Levitation System via Fuzzy PID Control Based on CS Algorithm

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Abstract— This study deals with the design of a magnetic levitation system based on PID type robust fuzzy logic controller (Fuzzy-PID) for improving system dynamics and stability. The proposed controller parameters are optimized with Cuckoo Search (CS) algorithm by minimizing a proposed objective function including the time domain response characteristics. The performance of the proposed controller is evaluated by means of extensive simulations for different conditions such as changes in references and load disturbance. Also, a comparison of this approach, CS based PID and CS based fractional order PID (FOPID) is introduced. The simulation results show that the CS based Fuzzy-PID controller has better performance in terms of overshoot, rise time, settling time and steady state error. Moreover, the disturbance rejection performance is investigated for the tuned controllers. The comparative results reveal that the proposed CS based Fuzzy-PID controller with least control effort performs better as compared to the PID and FOPID controllers.

Keywords—fractional order PID; PID; Fuzzy Logic Control; Magnetic ball levitation system; Cuckoo Search algorithm.

I. INTRODUCTION

The magnetic levitation (Maglev) system is based on the principle that a ferromagnetic object can resist gravity by means of a magnetic field produced by a coil which current flows through. Thus, the position of the levitated object can be actively changed and also the stiffness of the levitation system is investigated. Therefore, the magnetic levitation system has recently attracted extraordinary level of attention and also becomes growingly prevalent in industries. Accordingly, it has a wide variety of applications such as microelectromechanical systems [1], high-speed magnetic levitation trains [2], bearingless motors [3], vibration isolation [4], wind tunnel [5] and magnetic bearing [6].

PID controllers, are easy to design and implement, have been widely used in industrial applications for controlling nonlinear and linear systems [7]. On the other hand, the PID controllers cannot provide better response under the conditions of parameter uncertainties and external disturbances. Therefore, various approaches are proposed by many researchers to improve the performance of the PID controllers. FOPID controllers which are the expansions of the traditional PID

controllers based on the fractional calculus has been proposed by Podlubny [8]. FOPID control is nowadays a rapidly emerging control with proven performance in various applications such as automatic voltage regulator [9], wind energy system [10], motion and tracking control [11], designing aerospace control systems [12], varying time delay processes [13].

Recently, several efforts have been researched on nonlinear controller, mostly using fuzzy logic controller for nonlinear complex dynamical systems [14]. Accordingly, it has been revealed that the fuzzy logic controllers (FLCs) are used in various applications as providing robust controllers over the traditional controllers such as PID controller for the complex nonlinear and coupled systems, uncertain systems and the systems where no exact dynamics is required. Among different types of FLCs, like extensively utilized Fuzzy-PD, Fuzzy-PI, Fuzzy-PID in various systems, the fuzzy based PID controllers have recently become more common to overcome the nonlinear complex dynamical systems. In the literature, the structure of PID type fuzzy controller used in this work combines Fuzzy-PD and Fuzzy-PI controllers with the gains as the input scaling factors and the gains as the output scaling factors as described by [15-16].

The tuning parameters in Fuzzy-PID could be the selection of membership functions (MFs), input variables, scaling factors, rule base, fuzzification, inference and defuzzification methods. Various tuning methods have been presented in the literature for tuning the parameters of the Fuzzy-PID controllers such as particle swarm optimization (PSO) [17], Genetic Algorithm (GA) [18], Firefly Algorithm (FA) and Teaching Learning Based Optimization [19].

As one of the real industrial and versatile research-oriented systems, Maglev system displays a remarkably level of nonlinearities and uncertainties. Considering this from the control viewpoint, it causes some difficulties in designing a controller. Numerous control approaches have been proposed by various researchers to cope with the difficulties in controlling the position of the levitated object in Maglev system. PID controller has been applied to Maglev system by optimizing parameters using various methods [20-27]. To

achieve stable levitating and enhanced trajectory tracking control of Maglev system, the FOPID controller has been applied by various researchers [28-35]. Fuzzy control method is also applied some studies on Maglev system [36-37].

In this paper, a Fuzzy-PID controller tuned by CS algorithm is designed to achieve both satisfied tracking and disturbance rejection performance in Maglev system. The performance of the proposed Fuzzy-PID controller is compared with the PID and FOPID controllers tuned by CS. Moreover, the robustness of the Fuzzy-PID controller is investigated via the comparative analysis of results achieved from the tuned controllers.

This paper is organized as follows: in Section 2, the dynamic model of the Maglev system is presented. Section 3 comprises a brief overview of the CS algorithm. Section 4 covers the description of the Fuzzy-PID controller. In this section the design of other controllers is also given. Simulation results are presented and discussed in Section 5 and finally, a conclusion is presented in Section 6.

II. MODEL OF MAGNETIC LEVITATION SYSTEM

The Maglev system is based on the principle that a ferromagnetic object can resist gravity by means of a magnetic field produced by a coil which current flows through. In other words, the Maglev system is a ferromagnetic object suspended in a magnetic field which is controlled by the potential difference in the electric winding. As shown in Figure 1, the ferromagnetic object i.e. ball is moved by two forces: the gravity field and the electromagnetic force produced by the magnetic field of the coil.

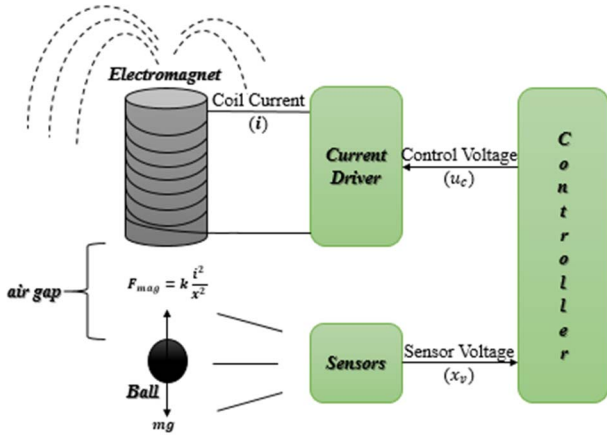


Fig. 1. Block diagram of the Maglev System

The simplest nonlinear model of the Maglev system related to the ball position x and the electromagnet coil current i is given by [25]:

$$m\ddot{x} = mg - k \frac{i^2}{x^2} \quad (1)$$

where m is the mass of the levitated object which is a ferromagnetic ball, g is the acceleration due to gravity, and k is an electromechanical conversion constant. This nonlinear

model of the Maglev system is linearized around the equilibrium point x_0 and i_0 :

$$\frac{\Delta X(s)}{\Delta I(s)} = \frac{-k_i}{s^2 - k_x} \quad (2)$$

where, $k_i = \frac{2g}{i_0}$, $k_x = \frac{2g}{x_0}$, and Δx and Δi are the small derivations from the equilibrium point x_0 and i_0 , respectively. Since the coil current i is proportional to the control voltage u , i.e., $i = k_1 u$, and the sensor output x_v is proportional to the position of the ball x , i.e., $x_v = k_2 x$, the transfer function from Δu to Δx_v is obtained as:

$$\frac{\Delta X_v(s)}{\Delta U(s)} = \frac{-k_1 k_2 k_i}{s^2 - k_x} \quad (3)$$

where k_1 is the control voltage to coil current gain and k_2 is the sensor gain. Finally, by substituting system parameters given in [25], reported in Table I, into Eq. 3:

$$G(s) = \frac{\Delta X_v(s)}{\Delta U(s)} = \frac{-3518.85}{s^2 - 2180} \quad (4)$$

TABLE I. IMPLEMENTATION PARAMETERS OF THE MAGLEV SYSTEM

Parameter	Notation	Value
Mass of the ball (kg)	m	0.02
Acceleration due to gravity (m/s²)	g	9.81
Equilibrium value of current (A)	i_0	0.8
Equilibrium value of position (m)	x_0	0.009
Control voltage to coil current gain (A/V)	k_1	1.05
Sensor gain (V/m)	k_2	143.48

III. CUCKOO SEARCH OPTIMIZATION ALGORITHM

Cuckoo Search (CS) optimization algorithm is inspired from the brood parasitic breeding strategy of certain species of cuckoos by laying their eggs in the nests of other host birds [38]. According to Cuckoo Search Algorithm, generating the new nest for cuckoos, the global random walk is performed by using a law named Lévy flights which is given below:

$$X_i(n+1) = X_i(n) + \alpha \otimes \text{levy}(\lambda) \quad (5)$$

where $n = 1, \dots, N$ indicates the current iterations in which N denotes the predetermined maximum iteration number. $\alpha > 0$ is the step size regarding the scales of the problem of interest. $\text{levy}(\lambda)$ represents Lévy flight process which is basically a random walk derived from the Lévy distribution with an infinite variance and infinite mean [38]:

$$\text{levy}(\lambda) = t^{-\lambda}, \quad (1 < \lambda \leq 3) \quad (6)$$

where t is the current iteration. The algorithm can also be extended to more complicated cases where each nest contains multiple eggs (a set of solutions) [38]. In that case, the new nest is randomly generated by using the following equation:

$$X_i^{new} = \begin{cases} X_i + \text{stepsize} \cdot \text{randn}, & \text{randn}_i > p_a \\ X_i, & \text{else} \end{cases} \quad (7)$$

where:

$$\text{stepsize} = 0.01 \cdot \left(\frac{\sigma(\beta) \cdot \text{randn}}{\text{randn}} \right)^{\frac{1}{\beta}} \cdot (X_i - X_{best}) \quad (8)$$

and randn is a random value in $[0, 1]$, β is a constant between $1 \leq \beta \leq 3$, and the standard deviation function $\sigma(\beta)$ is:

$$\sigma(\beta) = \left(\frac{\Gamma(1 + \beta) \cdot \sin(\pi \cdot \beta/2)}{\Gamma\left(\left(\frac{1 + \beta}{2}\right) \cdot \beta \cdot 2^{\left(\frac{\beta-1}{2}\right)}\right)} \right)^{\frac{1}{\beta}} \quad (9)$$

In light of the above, Cuckoo search optimization algorithm can be summarized as the following steps:

Step 1: Introduce X_i as a random population of n host nests.

Step 2: Generate a new solution X_i^{new} using by Lévy flights.

Step 3: Calculate its objective function: $J(X_i^{new})$.

Step 4: Select a nest randomly among the host nests say X_j and calculate its objective function value as $J(X_j)$.

Step 5: If $J(X_i^{new}) < J(X_j)$, then replace X_j by new solution X_{new} , else let X_j be the new solution.

Step 6: By using Lévy flights manner, leave a fraction of P_a of the worst nest by building new ones at new locations.

Step 7: Keep the current optimum nest,

Step 8: Go to Step 2, if max. iteration number is not reached.

Step 9: Find the optimum solution: X_i^{best} , $i = 1, \dots, n$.

IV. CONTROLLER DESIGN

A. Design of Fuzzy-PID Controller

The structure of the Fuzzy-PID controller used in this work is shown in Figure 2. This structure is a combination of Fuzzy-PD and Fuzzy-PI controllers with Se and Sce as input scaling factors and Su and Si as output scaling factors as in [15-16]. In this type of FLC structure, the inputs are the tracking error and derivative of the tracking error, and the FLC output is multiplied by Su and its integral multiplied by Si .

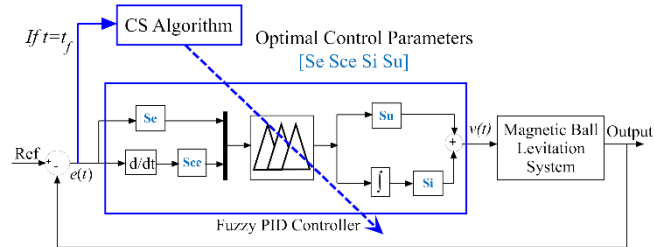


Fig. 2. Structure of Fuzzy-PID controller.

Both of the FLC input and output variables is represented with seven MFs namely NB (Negative Big), NM (Negative Medium), NS (Negative Small), ZE (Zero), PS (Positive Small), PM (Positive Medium), and PB (Positive Big) as shown in Figure 3. Gaussian membership function is used for both inputs and output of FLC, except NB and PB, owing to their remarkable advantages i.e. smooth functions, non-zero at all points and provides the actual information at all points. NB and PB are chosen as Z-shape and S-shape membership functions, respectively. The range of MFs is $[-1, 1]$ for both inputs and outputs. The fuzzy rule table is given in Table II, and also the fuzzy control surface is shown in Figure 4.

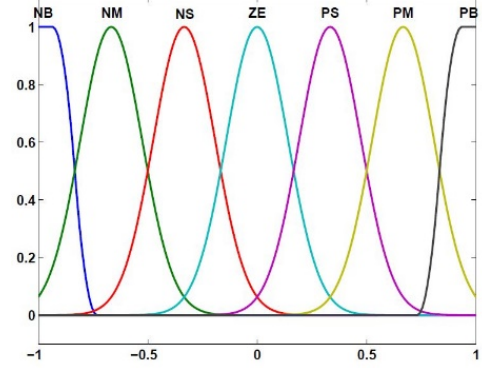


Fig. 3. Membership Functions of input and output variables.

TABLE II. FUZZY RULES TABLE

	$\hat{e}(t)$						
$e(t)$	NB	NM	NS	ZE	PS	PM	PB
NB	NB	NB	NB	NB	NM	NS	ZE
NM	NB	NB	NB	NM	NS	ZE	PS
NS	NB	NB	NM	NS	ZE	PS	PM
ZE	NB	NM	NS	ZE	PS	PM	PB
PS	NM	NS	ZE	PS	PM	PB	PB
PM	NS	ZE	PS	PM	PB	PB	PB
PB	ZE	PS	PM	PB	PB	PB	PB

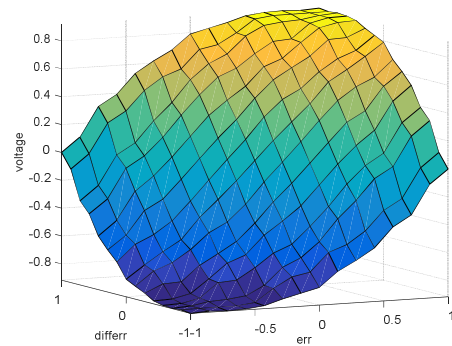


Fig. 4. Surface view of fuzzy inference.

B. Design of FOPID Controller

FOPID controller is characterized by five parameters. Compared to the traditional PID controller, FOPID has two additional parameters in which the orders of the derivative part μ and integral part λ are non-integer.

These extra parameters provide a greater flexibility in the design of controller and also can lead to achieve an improving dynamic performance. The transfer function of the FOPID is also given as:

$$C_{FOPID}(s) = K_p + K_i \frac{1}{s^\lambda} + K_d s^\mu \quad (10)$$

C. Design of Conventional PID Controller

A conventional PID controller's transfer function is given as:

$$C_{PID}(s) = K_p + K_i \frac{1}{s} + K_d s \quad (11)$$

D. Objective Function of the Optimization Algorithm

During the optimization, the objective function chosen here based on an objective function including time domain specifications which are overshoot (M_p), rise time (t_r), settling time (t_s), and steady state error (E_{ss}) is given as the following:

$$J = w_{Mp} M_p + w_{tr} t_r + w_{ts} t_s + w_{Ess} E_{ss} \quad (12)$$

In this work, several trials have been performed for the optimization process by using various values of the weighting factors (w_{Mp} , w_{tr} , w_{ts} , w_{Ess}). The values of the weights are $w_{Mp} = 0.9$, $w_{tr} = 100$, $w_{ts} = 0.9$, $w_{Ess} = 1000$. Those values are determined from these trials to improve the system response.

V. SIMULATION RESULTS

In this section, the PID, FOPID and Fuzzy-PID controllers are designed using the CS algorithm for the Maglev system to levitate the ball in the air-space.

TABLE III. PARAMETERS USED FOR TUNING THE CONTROLLERS

PID and FOPID Controllers	
Gains Range: $-5 \leq K_p \leq -1$, $-7.5 \leq K_i \leq -0.1$, $-0.2 \leq K_d \leq 0$	
$0 < \lambda$, $\mu \leq 2$	
The Parameters of Approximation $\omega_l = 0.001$, $\omega_h = 1000$, $N=5$	
FUZZY-PID Controller	
Gains Range:	$0 \leq S_e, S_{ce} \leq 2$ $-100 \leq S_i, S_u \leq 0$
Simulation Parameters	
Number of Iteration	$max_iter=100$
Number of trials	$R=10$
Simulation Time	$t=5s$
Input Parameters of CS	
Number of Nests	$n = 10$
Abandon Probability	$p_a = 0.25$
Constant in (6)	$\beta = 1.5$

In the simulation, all experiments have been performed by using MATLAB software on a MSI Laptop/Processor: Intel Core™ i7-6700 HQ CPU 2.6 GHz/Installed Memory (RAM):

32.00 GB/System Type: 64-bit win10 Operating System/VGA: GeForce NVidia GTX960M 2GB. Also, the tuning process of the controllers is conducted 10 runs with the given performance index. For carrying out the fractional order PID controller, Ninteger toolbox developed by Valério and Sá da Costa [39] has been used. As for the approximation algorithm, the Crone method (with $N=5$) proposed by Oustaloup [40] is used, which uses a recursive distribution of N poles and N zeros. The simulation parameters and the searching range of the controller parameters for the optimization are given in Table III.

The optimized scaling parameters of all the controllers applied to the Maglev system and best objective function values are reported in Table IV, according to the minimum value of each objective function among 10 runs.

TABLE IV. OPTIMIZED PARAMETERS AND PERFORMANCE INDEX VALUES OF THE PID, FOPID AND FUZZY-PID CONTROLLERS

FOPID and PID Controller Parameters					
Objective (J)	Kp	Ki	Kd	μ	λ
12.5453	-4.9089	-7.0825	-0.1668	1.0162	1.0051
15.4352	-4.8558	-7.0125	-0.0965	-	-
Fuzzy-PID Controller Parameters					
Objective (J)	Se	Sce	Si	Su	
1.1491	1.2569	0.0046	-98.2561	-95.7456	

For observing the transient-state and the steady-state performance of the optimized controllers, the step reference trajectory is applied to the Maglev system. The tracking performance of the designed controllers is shown in Figure 5 and the corresponding dynamic responses are listed in Table V.

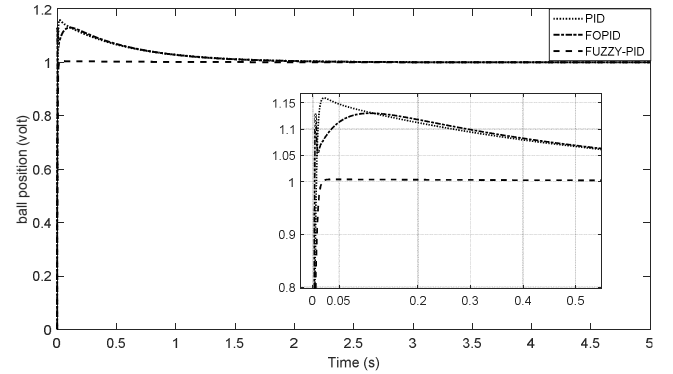


Fig. 5. Step responses of Maglev system with the tuned controllers using CS.

It can be easily seen from Figure 5 and Table V, the Maglev system exhibits better performance with the Fuzzy-PID controller tuned by CS as compared to the CS based PID and FOPID controllers. Consequently, the dynamic response characteristics obtained from the CS based Fuzzy-PID controller is closer to the dynamic response characteristics of the reference closed loop system in comparison with the other two controllers.

TABLE V. RESULTS OF RESPONSE CHARACTERISTICS OBTAINED FROM PID, FOPID AND FUZZY-PID CONTROLLERS BASED ON CS ALGORITHM

Controller	$M_p\%$	t_r 0.1 → 0.9	t_s ±5%	E_{ss}
PID	15.8937	0.0050	0.6690	0.000029
FOPID	13.0045	0.0020	0.6790	0.000030
Fuzzy-PID	0.4599	0.0070	0.0090	0.000027

The input signal is considered as a square wave with a mean of -1.5V and also applied to the Maglev system under a pulse magnitude of 0.5V at intervals of each 10s for the robustness test of the tuned controllers as shown Figure 6.

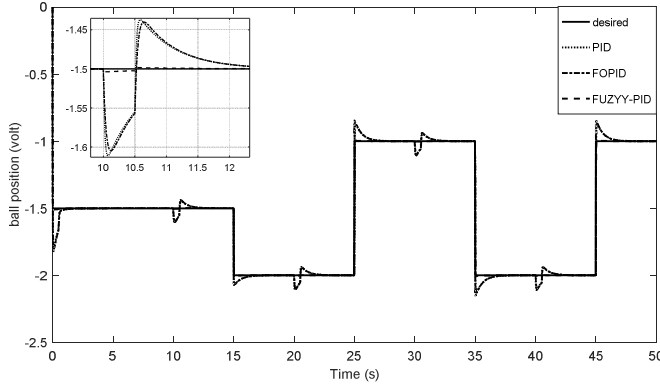


Fig. 6. Responses of Maglev system with the tuned controllers to periodic output disturbance.

The results from Figure 6 demonstrates that the Fuzzy-PID controller has the better dynamic performance and disturbance rejection ability as compared to the PID and FOPID controllers.

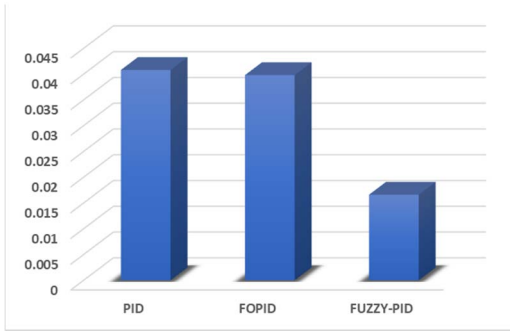


Fig. 7. Comparison of RMS values for the PID, FOPID and Fuzzy-PID controllers under disturbance.

Moreover, root mean square errors produced by the tuned controllers applied to the Maglev system are presented in Figure 7. These RMS values illustrates that the Fuzzy-PID controller can improve stability and dynamic response of the Maglev system under the disturbance.

It would again be useful to demonstrate the system under the different disturbance. Hence, in this paper, sinusoidal reference trajectory is selected as

$$r(t) = 5.8 + 1.8 \sin \frac{\pi}{2} t \quad (13)$$

The results are shown in Figure 8. It can be seen from the figure, the Fuzzy-PID controller has shorter settling time and

better trajectory tracking performance as compared to the others. Additionally, the statistical representation of variation of control input values for the trajectory tracking are given in Figure 9.

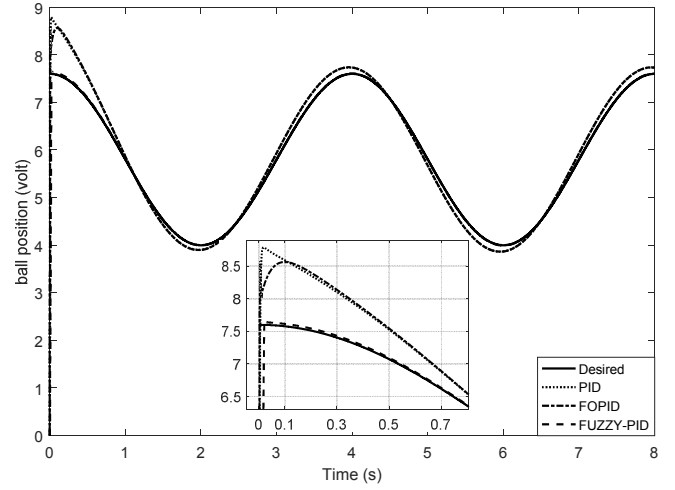


Fig. 8. Trajectory tracking performance for the PID, FOPID and Fuzzy-PID.

In Figure 9, it is seen that, the control effort of the Fuzzy-PID controller is significantly lower as compared to the PID and FOPID controllers. As a result, although all of the controllers almost produce the same average control signal, the best control performance is achieved by the Fuzzy-PID controller with least control effort.

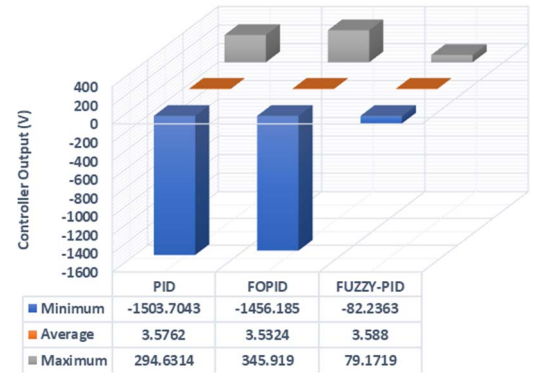


Fig. 9. Statistical analysis of the PID, FOPID and Fuzzy-PID controllers

VI. CONCLUSION

In this study, a simple and elegant approach for the design of the Fuzzy-PID controller tuned by the CS algorithm with respect to the proposed objective function is suggested for levitating the magnetic ball in the air space. The dynamic performance of the proposed controller is compared with the PID and FOPID controllers optimized by CS in terms of superior step response and trajectory tracking, better disturbance rejection and good control signal response. Based on comparison of results, it is revealed that the proposed CS tuned Fuzzy-PID controller exhibits superior dynamic response with least values of the overshoot, the rise time, the settling time and the steady state error as well as control effort. Consequently, the CS based

Fuzzy- PID controller is found effective and it shows robustness in improving the voltage stability under different reference input and a pulse signal with a periodic disturbance in the Maglev system.

REFERENCES

- [1] K. Komori and T. Yamane, "Magnetically levitated micro PM motors by two types of active magnetic bearings," *IEEE/ASME Transactions on Mechatronics*, vol. 6, no. 1, pp. 43-49, 2001.
- [2] M. Ono, S. Koga and H. Ohtsuki, "Japan's superconducting Maglev train," *IEEE Inst. & Measurement Mag.*, vol. 5, pp. 9-15, 2002.
- [3] F.-J. Lin, H.-J. Shieh and L.-T. Teng, "Hybrid controller with recurrent neural network for magnetic levitation system," *IEEE Transactions on Magnetics*, vol. 41, no. 7, pp. 2260-2269, 2005.
- [4] M. Tsuda, et al., "Vibration Transmission Characteristics Against Vertical Vibration in Magnetic Levitation Type HTS Seismic/Vibration Isolation Device," *IEEE Transactions on Applied Superconductivity*, vol. 19, no. 3, pp. 2249-2252, 2009.
- [5] G. Oćokoljić, B. Rašuo and M. Kozić, "Supporting system interference on aerodynamic characteristics of an aircraft model in a low-speed wind tunnel," *Aerospace Science and Technology*, vol. 64, pp. 133-146, 2017.
- [6] A. Mystkowski, A. Kierdelewicz, R. P. Jastrzebski, E. Dragašius and D. Eidukynas, "Flux measurement and conditioning system for heteropolar active magnetic bearing using Kapton-foil Hall sensors," *Mechanical Systems and Signal Processing*, vol. 115, pp. 394-404, 2019.
- [7] K. J. Åström and T. Häggglund, "The future of PID control," *Control Engineering Practice*, vol. 9, no. 11, pp. 1163-1175, 2001.
- [8] I. Podlubny, "Fractional-order systems and PID μ controller," *IEEE Transactions on Automatic Control*, vol. 44, pp. 208-214, 1999.
- [9] I. Pan and S. Das, "Frequency domain design of fractional order PID controller for AVR system using chaotic multi-objective optimization," *International Journal of Electrical Power & Energy Systems*, vol. 62, pp. 106-1018, 2013.
- [10] R. Melício, V. M. F. Mendes and J. P. S. Catalão, "Fractional-order control and simulation of wind energy systems with PMSG/full-power converter topology," *Energy Conversion and Management*, vol. 51, no. 6, pp. 1250-1258, 2010.
- [11] C. Muresan, S. Folea, G. Mois and E. Dulf, "Development and implementation of an FPGA based fractional order controller for a DC motor," *Mechatronics*, vol. 23, no. 7, pp. 798-804, 2013.
- [12] M. A. Aboelela, M. F. Ahmed and H. T. Dorrah, "Design of aerospace control systems using fractional PID controller," *Journal of Advanced Research*, vol. 3, no. 3, pp. 225-232, 2012.
- [13] C. I. Pop, C. Lonescu, R. D. Keyser and E. H. Dulf, "Robustness evaluation of fractional order control for varying time delay processes," *Signal, Image and Video Processing*, vol. 6, no. 3, pp. 453-461, 2012.
- [14] C. C. Lee, "Fuzzy logic in control systems: fuzzy logic controller, Part II," *IEEE Trans Syst Man Cybern*, vol. 20, pp. 404-18, 1990.
- [15] Z.-W. Woo, H.-Y. Chung, J.-J. Lin, "A PID type fuzzy controller with self-tuning scaling factors," *Fuzzy Sets Syst.* vol. 115, pp. 321-326, 2000.
- [16] E. Yesil, M. Guzelkaya, I. Eksin, "Self tuning fuzzy PID type load and frequency controller," *Energy Convers. Manage.*, vol. 45, no. 3, pp. 377-390, 2004.
- [17] B. K. Sahu, S. Pati and S. Panda, "Hybrid differential evolution particle swarm optimisation optimised fuzzy proportional-integral derivative controller for automatic generation control of interconnected power system," *IET Generation, Transmission & Distribution*, vol. 8, no. 11, pp. 1789-1800, 2014.
- [18] K. B. Yogesh, D. M. Hitesh, S. Houria, and B. Surekha, "Frequency Stabilization for Multi-area Thermal-Hydro Power System Using Genetic Algorithm-optimized Fuzzy Logic Controller in Deregulated Environment," *Electric Power Components and Systems*, vol. 43, no. 2, pp. 146-156, 2015.
- [19] S. Priyambada, B. K. Sahu and P. K. Mohanty, "Fuzzy-PID controller optimized TLBO approach on automatic voltage regulator," *International Conference on Energy, Power and Environment: Towards Sustainable Growth (ICEPE)*, Shillong, 2015, pp. 1-6, 2015.
- [20] I. Ahmad, M. Shahzad, P. Palensky, "Optimal PID control of magnetic levitation system using genetic algorithm," *IEEE international energy conference*, pp. 1429-1433, 2014.
- [21] S. Yadav, S. K. Verma, S. K. Nagar, "Optimized PID Controller for Magnetic Levitation System," *IFAC-PapersOnLine*, vol. 49, no. 1, pp. 778-782, 2016.
- [22] E. V. Kumar, J. Jerome, "LQR based optimal tuning of PID controller for trajectory tracking of magnetic levitation system," *Procedia Engineering*, vol. 64, pp. 254-264, 2013.
- [23] S. Yadav, S. K. Verma, and S. K. Nagar, "Performance enhancement of magnetic levitation system using teaching learning based optimization," *Alexandria Engineering Journal*, vol. 57, no. 4, pp. 2427-2433, 2017.
- [24] M. H. A. Yaseen and H. J. Abd, "Modeling and control for a magnetic levitation system based on SIMLAB platform in real time," *Results in Physics*, vol. 8, pp. 153-159, 2018.
- [25] A. Ghosh, et al., "Design and implementation of a 2-DOF PID compensation for magnetic levitation systems," *ISA Transactions*, vol. 53, no. 4, pp. 1216-1222, 2014.
- [26] D. Sain, "Real-Time implementation and performance analysis of robust 2-DOF PID controller for Maglev system using pole search technique," *Journal of Industrial Information Integration*, 2018.
- [27] M. Cetin and S. Iplikci, "A novel auto-tuning PID control mechanism for nonlinear systems," *ISA Transactions*, vol. 58, pp. 292-308, 2015.
- [28] S. K. Swain, D. Sain, S. K. Mishra, and S. Ghosh, "Real time implementation of fractional order PID controllers for a magnetic levitation plant," *AEU - International Journal of Electronics and Communications*, vol. 78, pp. 141-156, 2017.
- [29] S. Pandey, P. Dwivedi, and A. S. Junghare, "A novel 2-DOF fractional-order PID-D μ controller with inherent anti-windup capability for a magnetic levitation system," *AEU - International Journal of Electronics and Communications*, vol. 79, pp. 158-171, 2017.
- [30] G. Altintas, Y. Aydin, "Optimization of Fractional and Integer Order PID Parameters using Big Bang Big Crunch and Genetic Algorithms for a MAGLEV System," *IFAC-PapersOnLine*, vol. 50, pp. 4881-4886, 2017.
- [31] S. Folea, C. I. Muresan, R. De Keyser, and C. M. Ionescu, "Theoretical Analysis and Experimental Validation of a Simplified Fractional Order Controller for a Magnetic Levitation System," *IEEE Transactions on Control Systems Technology*, vol. 24, no. 2, pp. 756-763, 2016.
- [32] D. Sain, S. K. Swain, T. Kumar, and S. K. Mishra, "Robust 2-DOF FOPID Controller Design for Maglev System Using Jaya Algorithm," *IETE Journal of Research*, 2018.
- [33] P. Roy, M. Borah, L. Majhi, and N. Singh, "Design and implementation of FOPID controllers by PSO, GSA and PSOGSA for MagLev system," *2015 International Symposium on Advanced Computing and Communication (ISACC)*, Silchar, pp. 10-15, 2015.
- [34] S. K. Verma, S. Yadav, and S. K. Nagar, "Optimal fractional order PID controller for magnetic levitation system," *2015 39th National Systems Conference (NSC)*, Noida, pp. 1-5, 2015.
- [35] P. J. Gaidhane, A. Kumar, and M. J. Nigam, "Tuning of two-DOF-FOPID controller for magnetic levitation system: A multi-objective optimization approach," *6th Int. Conf. on Computer Applications In Electrical Engineering-Recent Advances (CERA)*, Roorkee, pp. 479-484, 2017.
- [36] J.J.H.-Casanias, M.A.M.-Vera, and B.D. B.-Hernandez, "Characterization and adaptive fuzzy model reference control for a magnetic levitation system," *Alexandria Engineering Journal*, vol. 55, pp. 2597-2607, 2016.
- [37] U. Sadeka, A. Sarjas, A. Chowdhury, and R. Sveclo, "Improved adaptive fuzzy backstepping control of a magnetic levitation system based on Symbiotic Organism Search," *Applied Soft Computing*, vol. 56, pp. 19-33, 2017.
- [38] X.-S. Yang and S. Deb, "Cuckoo search via Levy flights", in *Proc. of World Congress on Nature & Biologically Inspired Computing (NaBIC)* IEEE Publications, USA, pp. 210-214, 2009.
- [39] D. Valério, J. Sá da Costa, "Ninteger: a non-integer control toolbox for MatLab," *Proceedings of fractional differentiation and its applications*, Bordeaux, 2004.
- [40] A. Oustaloup, "La commande CRONE: commande robuste d'ordre non entiere," *Herme's*, Paris, 1991.